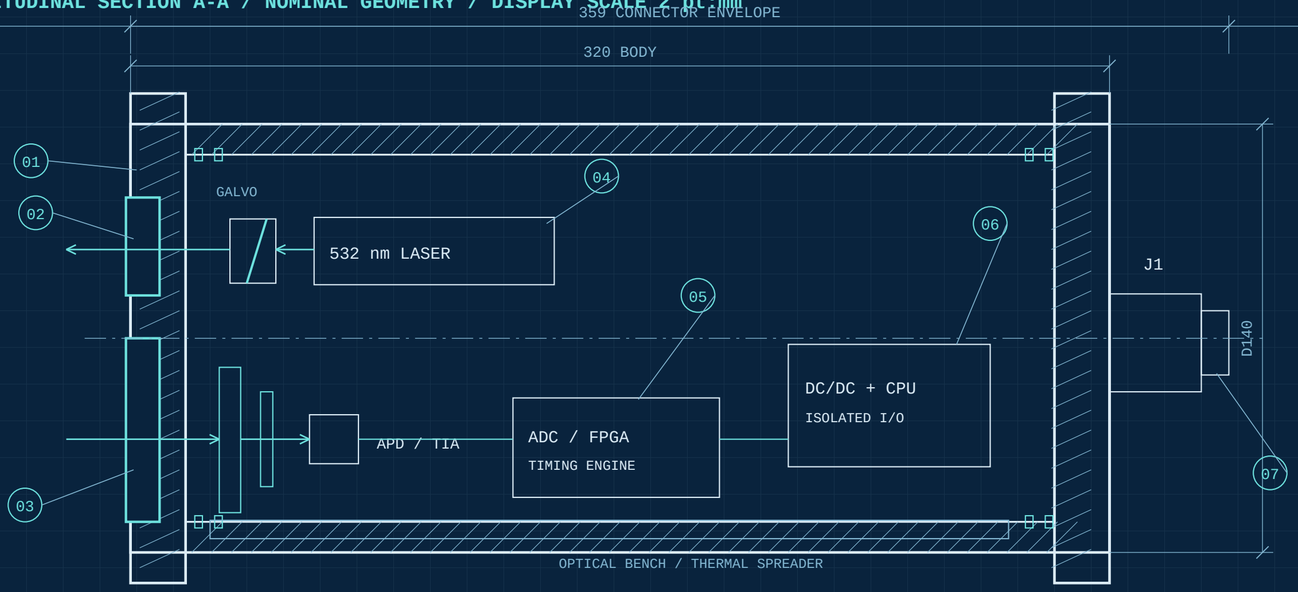
NEREID / 532

SUBSEA OPTICAL METROLOGY / CONCEPT ENGINEERING

GENERAL ARRANGEMENT & PRESSURE BOUNDARY

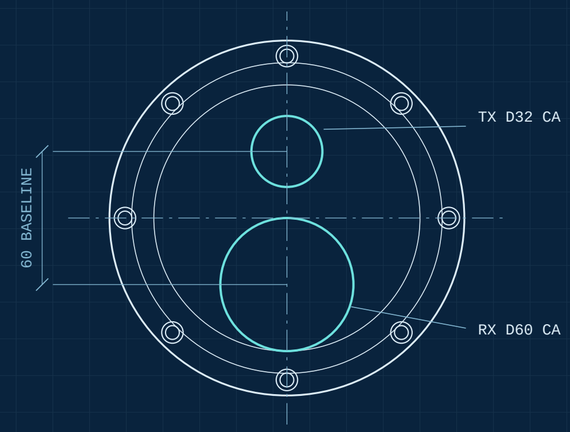
Orthographic views / longitudinal section / mounting interface / parts schedule

01 LONGITUDINAL SECTION A-A / NOMINAL GEOMETRY / DISPLAY SCALE 2 pt:mm



SECTION IS SCHEMATIC: component envelopes / retainers and wire routes not to fabrication detail.

02 FRONT ELEVATION / VIEW FROM WATER

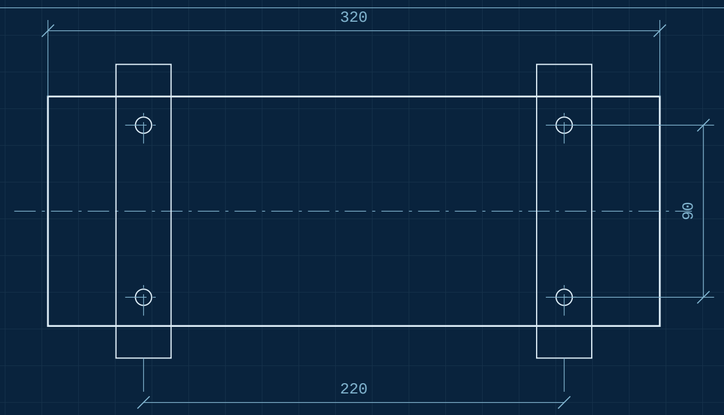


8 x M5 RETAINER FASTENERS ON D146 PCD

Front/rear flange D160; housing OD140 / ID120.

CA = clear aperture. Window blanks/seat design TBD.

03 UNDERSIDE / VEHICLE MOUNT DATUMS



4 x D8.5 THRU / 220 x 90 RECTANGULAR PATTERN

Clamp-on saddle rails; no holes through pressure tube.

04 ASSEMBLY SCHEDULE / CONCEPT MATERIAL SELECTION

ITEM	QTY	DESCRIPTION	MATERIAL / REMARK
01	1	Pressure tube + end flanges	Titanium Grade 5 / isolation at dissimilar metals
02	1	Transmit pressure window	Sapphire / D32 clear / thickness TBD by analysis
03	1	Receive pressure window	Sapphire / D60 clear / thickness TBD by analysis
04	1	Pulsed source + scan head	532 nm source / 1-axis galvo / monitor photodiode
05	1	Receiver + timing chain	Lens / filter / silicon APD / TIA / ADC / FPGA
06	1	Power + compute assembly	18-30 VDC / isolated converter / Ethernet
07	1	Rear wet-mate interface	12-contact concept / vendor and rating TBD
08	2	Mount saddles + seal sets	Titanium saddles / elastomer compound TBD

GENERAL NOTES 1. Depth target 300 m; geometry is not a pressure rating. 2. Window retention, seal glands, bolt preload and collapse margin require analysis.

3. Two radial body seals at each removable cap; inter-seal leak detection provision. 4. Dry nitrogen fill and humidity sensor proposed.

5. Deburr wetted edges; optical face protection required. Finish, tolerances, fastener torque and seal specifications remain unreleased.

6. Datum concept: A = front optical plane; B = housing axis; C = saddle clocking plane. Mounting dimensions are nominal interface targets.

CONCEPT ONLY / NOT RELEASED FOR MANUFACTURE

ALL DIMENSIONS mm. NUMERIC DIMENSIONS GOVERN. DO NOT SCALE PDF.

Assumed geometry; pressure, optics, thermal performance and laser safety require validation.

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SHEET 1 OF 3

ENGINEERING CONCEPT

NO TEST DATA

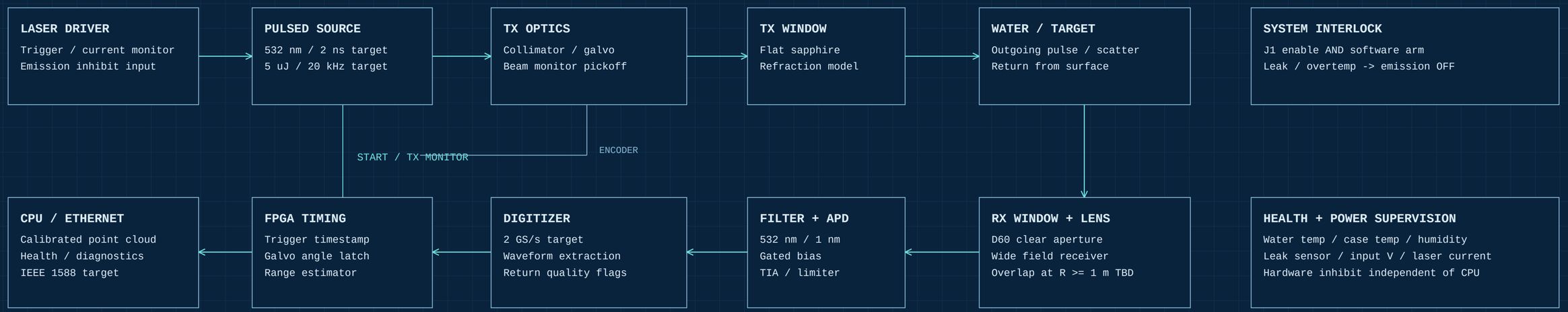
NEREID / 532

SUBSEA OPTICAL METROLOGY / CONCEPT ENGINEERING

OPTICAL, ELECTRICAL & TIMING ARCHITECTURE

Functional schematic / power domains / logical connector allocation / scan geometry

01 SIGNAL PATH / BLOCK SCHEMATIC - NOT A COMPONENT-LEVEL CIRCUIT



ACCENT ARROWS: optical / sampled signal flow. All modules inside dry pressure boundary except water path.

Beam monitor measures emission timing; per-pulse reference corrects driver delay. Receive gate suppresses early backscatter.

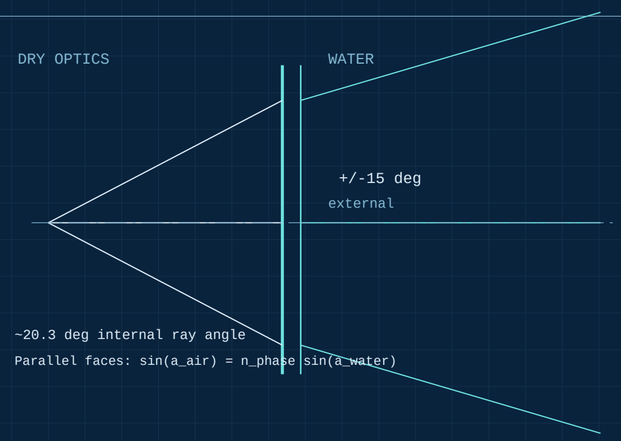
02 POWER & LOGICAL J1 ALLOCATION

J1 18-30 VDC -> surge/reverse protection -> EMI filter
-> isolated DC/DC -> laser rail / analog rail / digital rail
Chassis bond separate from isolated signal return.

CONTACTS	FUNCTION	IMPLEMENTATION NOTE
1 / 2	VIN+ / VIN-	24 V nominal; 60 W peak budget
3 / 4	ETH TX+ / TX-	100BASE-TX differential pair
5 / 6	ETH RX+ / RX-	100BASE-TX differential pair
7 / 8	PPS+ / PPS-	Isolated differential timing input
9 / 10	ENABLE+ / ENABLE-	Isolated laser permit input
11 / 12	SERVICE+ / SERVICE-	Isolated RS-485 / maintenance

Logical assignment only; connector pin geometry/rating not selected.

03 REFRACTED SCAN / BISTATIC RANGE MODEL



FOV target: 30 deg in water
Flat port / 1-axis fan scan
RX full FOV target >= 40 deg
Vehicle motion supplies axis 2

Optical path L = c0 * dt / n_group
L = |P - TX| + |P - RX|
Solve on calibrated TX ray.
R ~ L/2 only in far field.

n_phase for refraction;
n_group for flight time.

TIMING BUDGET 20 kHz pulse rate -> 50 us pulse interval. 1,000 angular samples/line -> 20 lines/s before turnaround and rejected returns.

2 GS/s sampling -> 0.5 ns intervals -> ~56 mm monostatic range spacing at assumed n_group = 1.34; interpolation is required for a 20 mm precision target.

Flat-port and 60 mm bistatic baseline calibration are mandatory. Range gate 1-15 m corresponds to ~8.9-134 ns monostatic-equivalent delay.

LASER CONTROL Classification and accessible-emission assessment pending. Hardware permit, watchdog and fail-off driver are required design provisions.

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SHEET 2 OF 3
ENGINEERING CONCEPT
NO TEST DATA

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SUBSEA OPTICAL METROLOGY / CONCEPT ENGINEERING

SENSOR DATA & VERIFICATION BASIS

Separate data sheet / assumed design targets / derived quantities / validation plan

01 SENSOR SPECIFICATION / ALL VALUES ARE CONCEPT TARGETS

PARAMETER	TARGET / DESIGN ASSUMPTION
Principle	Pulsed direct time of flight; bistatic receiver
Wavelength / pulse width	532 nm / 2 ns nominal
Pulse energy / rate	5 uJ / 20 kHz nominal
Optical average power	0.10 W, derived before optical losses
Range objective	1-15 m in clear water; not demonstrated
Precision objective	<=20 mm, 1 sigma at 5 m under test conditions
Absolute accuracy	Unspecified until calibration and tank test
Scan field / angular step	30 deg in water / nominal 0.03 deg
Scan throughput	20,000 shots/s; <=20 lines/s at 1,000 shots/line
Receive aperture / FOV	D60 clear window / >=40 deg full optical FOV
Receiver / digitizer	Silicon APD + TIA / 2 GS/s target
Power / electrical	18-30 VDC; 35 W nominal / 60 W peak budget
Data / timing	100BASE-TX / PPS / PTP support target
Depth / temperature	300 m target / water 0 to +35 C target
Mechanical envelope	D160 max x 359 incl. connector; 320 body
Mass / buoyancy	TBD by CAD mass model; no claimed value
Output	XYZ, intensity, time, scan angle, quality flags

02 CALCULATIONS / NOT MEASURED SENSOR DATA

A TIME OF FLIGHT / n_group = 1.34 assumed
c0 = 299,792,458 m/s; v_group ~ 223.73 x 10^6 m/s
At 5 m: dt ~ 44.70 ns (monostatic equivalent)
20 mm range change -> ~179 ps timing change

B SAMPLING / SCAN GEOMETRY
30 deg / 1,000 samples ~ 0.03 deg per shot
At 5 m: angular pitch ~ 2.62 mm near boresight
Swath at 5 m: 2R tan(15 deg) ~ 2.68 m
Beam divergence target 2 mrad -> ~10 mm growth at 5 m
Sample pitch does not equal spatial resolution.

C EXTERNAL PRESSURE / seawater density 1,025 kg/m^3
p_gauge = rho*g*h ~ 3.02 MPa at 300 m (g=9.81)
On D60 clear aperture: F ~ p*A = 8.53 kN
Window seat load also depends on unsupported diameter.
No collapse, fatigue, seal or window stress analysis done.

D ATTENUATION EXAMPLE / homogeneous water only
Two-way transmission factor ~ exp(-2*c_att*R)
At 10 m: c_att=0.10/m -> 0.135; 0.30/m -> 0.00248
Excludes geometric loss, reflectance and receiver noise.
No link-budget closure or range guarantee is implied.

03 OUTPUT RECORD / PROPOSED WIRE FORMAT

FIELD	TYPE / UNIT	MEANING
timestamp_ns	uint64 / ns	TAI epoch; validity flag required
xyz_sensor	3 x float32 / m	Right-handed sensor coordinate frame
range_tx	float32 / m	Distance along calibrated transmitted ray
intensity	uint16 / count	Uncalibrated amplitude; saturation flagged
angle / line_id	float32 rad / uint32	Water-side ray angle / line index
quality/status/pad	3 x uint16	Quality, health flags, reserved zero pad

Frame: +Z forward along optical axis; +X starboard; +Y down.
Packed point record: 40 bytes; 20k records/s -> 0.8 MB/s payload.
Add packet framing, sequence, CRC and calibration ID at transport layer.
No-return records carry validity flag; invalid coordinates use NaN.
Health stream: case/water temp, humidity, input V/I, interlock state.

04 ACCEPTANCE PLAN / UNRESOLVED ENGINEERING

V1 OPTICS / Ray trace through sapphire and seawater; verify full scan clearance, APD etendue, FOV overlap and internal scatter.

V2 RANGE / Tank target at 1, 5, 10, 15 m; record reflectance, attenuation, ambient light, water T/S and n_group model. >=1,000 valid returns/position; report bias, sigma and dropout.

V3 PRESSURE / External-pressure FEA, windows, glands and preload; establish proof protocol and fatigue life before hydrotest.

V4 THERMAL / Test 35 W continuous and 60 W peak duty in still water; verify source drift, APD noise and maximum case temperature.

V5 INTEGRATION / Timestamp alignment, EMC, leakage, corrosion, connector service life and hardware fail-off fault injection.

V6 SAFETY / Determine laser class for every accessible condition; release interlock design only after documented assessment.

REFERENCE BASIS: [R1] Non-coaxial underwater lidar research (2023) [R2] NIST speed of light. Targets above are original concept assumptions.

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SHEET 3 OF 3
ENGINEERING CONCEPT
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